

Trial Scale Protein Extraction

Overview

Evaluate protein extraction conditions on a small scale before proceeding to large-scale production.

Harvest the supernatant

A. Materials

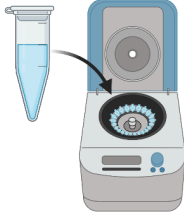
Item	Amount
Liquid culture of <i>B. subtilis</i>	Small scale, 5–50 mL
Centrifuge tubes	Sized to culture volume
Centrifuge	1
EDTA-free protease inhibitor	As required (follow product dilution)
Fresh clean tubes for supernatant	As required
Ice bucket / 4°C access	As required

B. Procedure

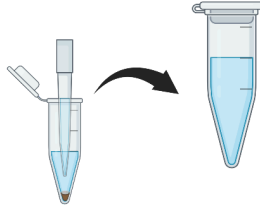
1. Centrifuge at 4,000 - 8,000 × g for 10-20 min at 4°C to separate the cell pellet and supernatant.
2. Retain the supernatant and cell pellet for later use.
3. Immediately add EDTA-free protease inhibitor to the supernatant.
4. Clarify the supernatant further by a second centrifugation (e.g., 15,000 × g, 15–20 minutes, 4°C)

Trial Scale Extraction

- ① Pellet the induced liquid culture by centrifugation at $4,000\text{--}8,000 \times g$ for 10–20 minutes at 4°C .



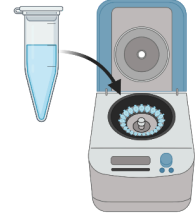
- ② Keep the supernatant and discard the cell pellet



- ③ Immediately add EDTA-free protease inhibitor to the supernatant.



- ④ Clarify the supernatant further by a second centrifugation $15,000 \times g$, 15–20 minutes, 4°C



Harvest the cell pellet

A. Materials

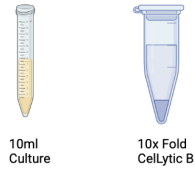
Item	Amount
Liquid culture of <i>B. subtilis</i>	Small scale, 5–50 mL
Centrifuge tubes	Sized to culture volume
Centrifuge	1
CellLytic B 10X concentrate	10-fold
EDTA-free protease inhibitor	As required (follow product dilution)
Fresh clean tubes for supernatant	As required
Ice bucket / 4°C access	As required
Vortex	1

B. Procedure

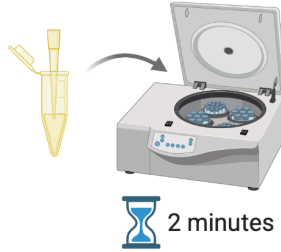
1. Prepare 10-fold dilution of CellLytic B 10X into your chosen buffer.
2. Grow a 10 mL culture of your *B. subtilis* strain under normal induction conditions.
3. Take 1.5 mL of culture (OD₆₀₀ 0.5–1.0) per dilution condition, and centrifuge at full speed for 2 minutes to pellet the cells. Prepare one tube per CellLytic B dilution
4. Remove and discard the medium, then resuspend each cell pellet in 0.4 mL of a different CellLytic B dilution.
5. Vortex briefly, then mix for 5–10 minutes at room temperature to let the detergent extract soluble proteins.
6. Centrifuge at full speed for 5 minutes to pellet insoluble material.
7. Carefully separate the supernatant (soluble protein fraction) from the pellet (insoluble/inclusion body fraction).

Harvesting Cell Pellet

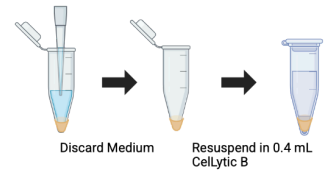
- 1 Grow 10 mL culture and prepare 10× CellLytic B dilution



- 2 1.5 mL culture, centrifuge full speed 2 minutes



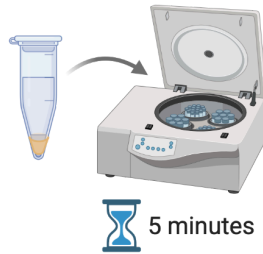
- 3 Discard medium, resuspend pellet in 0.4 mL CellLytic B dilution



- 4 Vortex briefly, and mix 5–10 min at room temperature



- 5 Centrifuge full speed 5 min to pellet insoluble material



- 6 Carefully separate soluble fraction from insoluble pellet

